

Composing Functions from Symbolic Rules

Answer Key

Algebra 2 Functions

Quick Answers

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|--------------------|------------------------------|
| 1. 10 | 6. $\frac{1}{x}$ |
| 2. 1, 9, 25 | 7. $2x^2 + 1, 4x^2 + 4x + 1$ |
| 3. $x^2 - 6x + 10$ | 8. 5 |
| 4. $x^2 - 2$ | 9. x, x |
| 5. $x^2 + 8x + 15$ | 10. — |
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Curriculum

Level: Algebra 2 Functions

HSF-BF.A.1c – problems 1, 2, 3, 4, 5, 6, 7, 8, 10

HSF-BF.B.4 – problem 9

Difficulty 1–4 of 5 across 14 tagged checks.

1. $(f \circ g)(2)$ for $f(x) = x + 4$, $g(x) = 3x$.

Step 1. Identify which rule is inner. In $f \circ g$ the letter closest to the input is g , so g runs first.

Step 2. $g(2) = 3 \cdot 2 = 6$.

Step 3. Feed that output to f : $f(6) = 6 + 4 = 10$.

Step 4. Same result from the composed rule: $f(g(x)) = 3x + 4$, and at $x = 2$ that is $6 + 4$.

$$(f \circ g)(2) = 10$$

Common wrong answer: 14 — the rules were run in the other order, $g(f(2)) = 3(2 + 4)$. The inner rule is the one written next to the input.

2. Table of $(g \circ f)(x)$ for $f(x) = 2x - 1$, $g(x) = x^2$.

Step 1. f is inner here, so each input is doubled and reduced by one first; the square is applied to that result, never to the original input.

Step 2. Middle column: $f(0) = -1$, $f(2) = 3$, $f(3) = 5$.

Step 3. Last column, squaring each of those:

$$g(f(0)) = (-1)^2 = \boxed{1} \quad g(f(2)) = 3^2 = \boxed{9} \quad g(f(3)) = 5^2 = \boxed{25}$$

Teaching note: the composed rule is $(g \circ f)(x) = (2x - 1)^2$. Reading the finished table left to right is exactly that rule, one input at a time — worth pointing out, because the next problems ask for the rule directly.

3. $(f \circ g)(x)$ for $f(x) = x^2 + 1$, $g(x) = x - 3$.

Step 1. g is inner. Wherever f shows its input, write the whole inner rule in parentheses:
 $f(g(x)) = (x - 3)^2 + 1$.

Step 2. Expand the square: $(x - 3)^2 = x^2 - 6x + 9$.

Step 3. Add the +1: $x^2 - 6x + 9 + 1 = x^2 - 6x + 10$.

$$(f \circ g)(x) = x^2 - 6x + 10$$

Common wrong answer: $x^2 + 9 + 1 = x^2 + 10$ — the square of a binomial was taken term by term. $(x - 3)^2$ is $(x - 3)(x - 3)$, which has a middle term.

4. $(g \circ f)(x)$ for the same $f(x) = x^2 + 1$, $g(x) = x - 3$.

Step 1. Now f is inner, so the input of g becomes $x^2 + 1$.

Step 2. g subtracts 3 from its input: $(x^2 + 1) - 3$.

Step 3. Combine the constants: $1 - 3 = -2$.

$$(g \circ f)(x) = x^2 - 2$$

Compare with problem 3. The same two rules gave $x^2 - 6x + 10$ one way and $x^2 - 2$ the other. Composition is not commutative — ask the student to check one input, say $x = 1$: 4 versus -1 .

5. $(f \circ g)(x)$ for $f(x) = x^2 - 2x$, $g(x) = x + 5$.

Step 1. f 's input appears twice, in x^2 and in $-2x$, so the inner rule is substituted in both places: $f(g(x)) = (x + 5)^2 - 2(x + 5)$.

Step 2. Expand each piece: $(x + 5)^2 = x^2 + 10x + 25$ and $-2(x + 5) = -2x - 10$.

Step 3. Combine: $x^2 + 10x + 25 - 2x - 10 = x^2 + 8x + 15$.

$$(f \circ g)(x) = x^2 + 8x + 15$$

Common wrong answer: $x^2 + 10x + 25 - 2x$ — the distribution stopped after the first term of $-2(x + 5)$.

6. $f(g(x))$ for $f(x) = \frac{1}{x-2}$, $g(x) = x + 2$.

Step 1. The input of f is the whole inner rule, so the denominator becomes $(x + 2) - 2$, not $x - 2$.

Step 2. Simplify that denominator: $(x + 2) - 2 = x$.

Step 3. The composition is therefore one over x .

$$f(g(x)) = \frac{1}{x}$$

Teaching note: the shift g applies is undone by the shift built into f , which is a first look at why the inverse test in problem 9 works. Worth saying aloud: $x = 0$ must still be excluded, because $g(0) = 2$ is exactly the input f cannot take.

7. Both orders for $f(x) = 2x + 1$, $g(x) = x^2$.

Step 1. $(f \circ g)(x)$: the input of f is x^2 , so double it and add one.

Step 2. $f(g(x)) = 2(x^2) + 1$.

$$(f \circ g)(x) = 2x^2 + 1$$

Step 3. $(g \circ f)(x)$: now the input of g is $2x + 1$, so that whole binomial gets squared:
 $(2x + 1)^2 = 4x^2 + 4x + 1$.

$$(g \circ f)(x) = 4x^2 + 4x + 1$$

Step 4. The two rules differ — swapping the order changes the function, so composition is not commutative.

Common wrong answer: $4x^2 + 1$ for the second one — the middle term $4x$ is lost when $(2x + 1)^2$ is squared term by term.

8. Solve $(f \circ g)(x) = 12$ **for** $f(x) = x + 3$, $g(x) = 2x - 1$.

Step 1. Build the composed rule first: $f(g(x)) = (2x - 1) + 3$, which simplifies to $2x + 2$.

Step 2. Set that rule equal to the target value: $2x + 2 = 12$.

Step 3. Subtract 2 from both sides: $2x = 10$; divide by 2.

Step 4. Check by running the input through both machines: $g(5) = 9$ and $f(9) = 12$ ✓

$$x = 5$$

Common wrong answer: $x = 6.5$ — the equation $2x - 1 = 12$ was solved, i.e. the outer rule was never applied.

9. Inverse test for $f(x) = 3x + 7$, $g(x) = \frac{x-7}{3}$.

Step 1. A composition test needs both directions, because a rule can undo another on one side only. Do $f(g(x))$ first.

Step 2. $f(g(x)) = 3\left(\frac{x-7}{3}\right) + 7$. The factor of 3 cancels the division, leaving $(x - 7) + 7$.

$$f(g(x)) = x$$

Step 3. Now the other direction: $g(f(x)) = \frac{(3x + 7) - 7}{3} = \frac{3x}{3}$.

$$g(f(x)) = x$$

Step 4. Both compositions return the input untouched, so f and g are inverses: $g = f^{-1}$.

Listen for: “I only did one side.” One direction alone is not enough in general — the two-sided check is what the definition asks for.

10. Decomposing $h(x) = \frac{1}{x^2+6}$. *Open response — many correct answers, flagged for manual review. The response below is a model full-credit answer.*

Step 1. Look for a chunk of h that is computed first when you evaluate it by hand. Given a number, you would square it, add 6, and only then divide into 1. That last-computed step belongs to the outer rule.

Step 2. A clean split is $g(x) = x^2 + 6$ (the inside chunk) and $f(x) = \frac{1}{x}$ (the outside chunk). Neither is the identity rule.

Step 3. Compose back to check: $f(g(x)) = \frac{1}{g(x)} = \frac{1}{x^2+6} = h(x) \checkmark$

Step 4. Other splits also earn full credit, for example $g(x) = x^2$ with $f(x) = \frac{1}{x+6}$, which composes to $\frac{1}{x^2+6}$ as well.

Full credit: two rules are named, neither is the identity, and the student composes them back and lands on h . A pair that is merely asserted, with no composition shown, is missing the justification the question asks for.